Xiangyu Wang

Mathematics Ph.D. Candidate | B.S. in Computer Science
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RESEARCH PROFILE

Proof-oriented mathematician with eight research papers and preprints (six sole-authored), including an accepted journal article. Expertise in complex mathematical arguments, multiscale analysis, sharp examples, and identifying limitations of proof techniques. Seeking to apply this background to AI mathematical reasoning, rigorous proof evaluation, and AI-driven mathematical discovery.

EDUCATION & GRADUATE TRAINING

University of Illinois Urbana-Champaign 2023–present

Ph.D. in Mathematics (in progress) | Advisor: Xiaochun Li

Graduate coursework: CS 542 · Statistical Reinforcement Learning (theory, sample complexity, online/offline RL, off-policy evaluation, policy gradients)

Boston University 2017–2020

B.S. in Computer Science

SELECTED RESEARCH CONTRIBUTIONS

- **Fourier restriction / Kakeya.** Developed restriction estimates using multilinear Kakeya inequalities, improving the range obtained by direct polynomial-partitioning methods in sufficiently high dimensions.
- **Local smoothing / wave packets.** Refined wave-packet density arguments for wave equations and constructed a counterexample exposing limitations of an existing proof framework.
- **Weighted restriction / multiscale methods.** Studied cone and planar weighted restriction using refined decoupling, geometric incidence methods, and scale-dependent arguments.

TECHNICAL BACKGROUND & RESEARCH INTERESTS

Programming: Python; PyTorch (coursework exposure).

Theoretical foundations: Statistical reinforcement learning; probability; rigorous proof construction; geometric and multiscale analysis.

AI interests: Mathematical reasoning, theorem proving, verification of long proofs, and research-level model evaluation.

PUBLICATIONS & PREPRINTS

1. *Restriction Estimates Using Multilinear Kakeya* · arXiv:2610.17752
2. *L^p-weighted Cone Restriction Estimates* · arXiv:2610.16294
3. *Wave Packet Density in Local Smoothing: Refinements and Limitations* · arXiv:2609.34675
4. *A Weighted Restriction Estimate in $\mathbb{R}^2$* · arXiv:2609.12587 · with Zhenbin Cao, Feilong Guo, and Junfeng Li
5. *On Oscillatory Integral Operators satisfying the Cinematic Curvature Condition* · arXiv:2512.22516
6. *On Cone Restriction Estimates in Higher Dimensions* · arXiv:2411.02961 · **accepted, Journal of Fourier Analysis and Applications**
7. *On Integers whose Friable Component is Large* · arXiv:2609.37507 · with Gérald Tenenbaum
8. *A Short Proof of a Theorem of Pan and Zhang* · arXiv:2609.37508

TEACHING & MENTORING

University of Illinois Urbana-Champaign. MATH 241 (Calculus III) mathematics tutor; grader for Algebraic Geometry (MATH 512), Real Analysis (MATH 444), and Complex Variables (MATH 448). Scheduled to lead discussion sections in Spring 2027.